

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)

Question Count: 5

Total Marks: 25

Code of Conduct

I BAIRAGANI DINESH -192372189 (Name / Reg No) certify that this submission is my original work and that I

have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: BAIRAGANI DINESH

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing

Definition of Cloud Computing

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet. Instead of purchasing and maintaining physical hardware, users can access these resources on demand and pay only for what they use.

Characteristics of Cloud Computing

1. On-Demand Self-Service

Users can create and manage computing resources such as virtual machines and storage without needing assistance from the service provider.

2. Broad Network Access

Cloud services are available over the internet and can be accessed from desktops, laptops, smartphones, and tablets from anywhere with an internet connection.

3. Resource Pooling

Cloud providers use shared infrastructure to serve multiple customers. Resources are allocated dynamically based on user demand, improving efficiency and reducing costs.

4. Rapid Elasticity (Scalability)

Resources can be increased or decreased quickly depending on workload. This allows businesses to handle traffic spikes without buying additional hardware.

5. Measured Service

Cloud providers monitor resource usage such as storage, bandwidth, and processing power. Customers pay only for the services they actually use, making cloud computing cost-effective.

Difference from Traditional On-Premise Computing

No need to purchase and maintain expensive hardware.

Resources can be accessed from anywhere through the internet.

Easy to scale resources up or down based on demand.

Lower maintenance costs and faster deployment of applications.

Conclusion

Cloud computing provides flexible, scalable, and cost-effective IT services over the internet. Its features such as on-demand access, resource sharing, scalability, and pay-as-you-go pricing make it more efficient than traditional on-premise computing.

2.Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Trace the Evolution of Cloud Computing

Cloud computing evolved through several stages of technological development. Advances in hardware, networking, distributed systems, and internet software made modern cloud platforms possible.

1. Hardware Evolution (1960s–1980s)

Mainframe computers were used to serve multiple users.

Virtualization technology allowed one physical server to run multiple virtual machines.

Improved processors, storage, and networking increased computing performance.

2. Internet Evolution (1990s)

The widespread adoption of the Internet enabled computers to communicate globally.

Faster broadband connections made online services more reliable.

Web browsers allowed users to access applications from anywhere.

3. Internet Software Evolution (2000s)

Web applications replaced traditional desktop software.

Technologies such as web services, APIs, and Service-Oriented Architecture (SOA) enabled software integration.

Companies started delivering software through the internet instead of installing it on local computers.

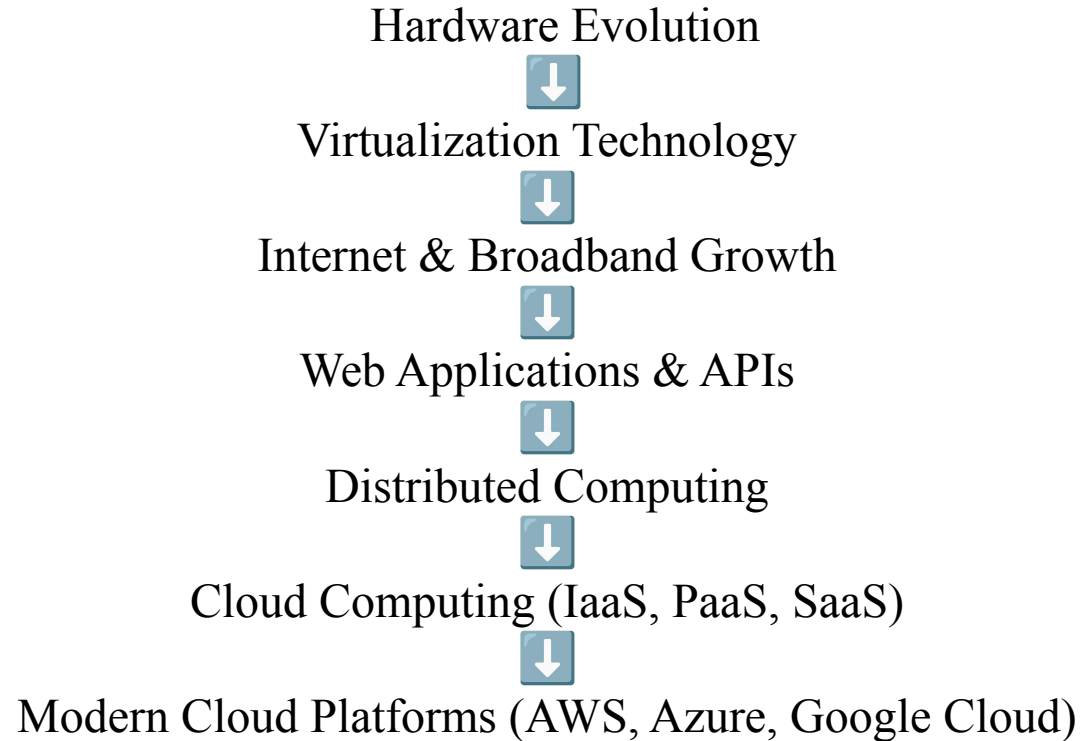
4. Modern Cloud Platforms (2010s–Present)

Cloud providers offer Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Features such as virtualization, containerization, automation, and elastic scaling improve efficiency.

Users can access secure, scalable, and cost-effective computing resources on demand.

Evolution Flow:-



Conclusion:

The evolution from powerful hardware and virtualization to internet-based software transformed computing into cloud platforms. Today, cloud computing provides scalable, reliable, and on-demand services that support businesses, education, and modern digital applications.

Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Introduction

Server virtualization is a technology that allows a single physical server to run multiple virtual servers called **Virtual Machines (VMs)**. Each VM has its own operating system and applications, making better use of hardware resources.

Role of Server Virtualization

Improves Resource Utilization – Multiple VMs share the same physical server, reducing idle resources.

Reduces Cost – Lowers hardware, electricity, and maintenance expenses.

Supports Scalability – New virtual servers can be created quickly as demand increases.

Provides Isolation – Each VM works independently, so problems in one VM do not affect others.

Simplifies Management – Virtual machines can be easily backed up, migrated, and restored

Virtualization

A technology that creates multiple virtual machines on one server.

Improves hardware utilization.

Acts as the foundation of cloud computing.

Cloud Computing

A service model that delivers computing resources over the internet.

Provides on-demand services like IaaS, PaaS, and SaaS.

Uses virtualization to offer scalable cloud services.

Discuss the role of data centers in delivering cloud services..

Role of Data Centers in Delivering Cloud Services

A **data center** is a facility that contains servers, storage devices, networking equipment, and cooling systems. It is the backbone of cloud computing and delivers cloud services to users over the internet.

Role of Data Centers

Store and manage large amounts of data.

Host cloud applications and services.

Provide high availability and reliable performance.

Ensure data security through backup and disaster recovery.

Support scalability by adding resources when demand increases.

Conclusion

Data centers are the foundation of cloud computing. They provide secure, reliable, and scalable infrastructure that enables users to access cloud services anytime and from anywhere.

Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Service Model	Description	Real-World Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage, and networking over the internet.	Amazon Web Services EC2
PaaS (Platform as a Service)	Provides a platform to develop, test, and deploy applications.	Google Cloud App Engine
SaaS (Software as a Service)	Delivers ready-to-use software through a web browser.	Google Gmail
XaaS (Anything as a Service)	Provides any IT service over the internet, such as storage, security, or databases.	Dropbox (Storage as a Service)

Conclusion:-

- **IaaS** provides infrastructure.
- **PaaS** provides a development platform.
- **SaaS** provides ready-to-use software.
- **XaaS** refers to any service delivered through the cloud.